Supply Chain Control Tower Data Analysis Report

Executive Summary:

This report analyzes supply chain operations data to identify key patterns in sales, inventory, supplier performance, and pricing.

Key Findings:

- Demand Forecast is the strongest driver of sales
- Inventory is not aligned with actual demand
- Supplier performance is inconsistent
- Dataset contains critical data quality issues

Business Objective

- The goal of this analysis is to:
- Understand demand patterns
- Evaluate inventory efficiency
- Identify supply chain risks
- Detect data inconsistencies
- Prepare dataset for advanced analytics

Dataset Overview

- Total Records: 91,250
- Total Features: 15

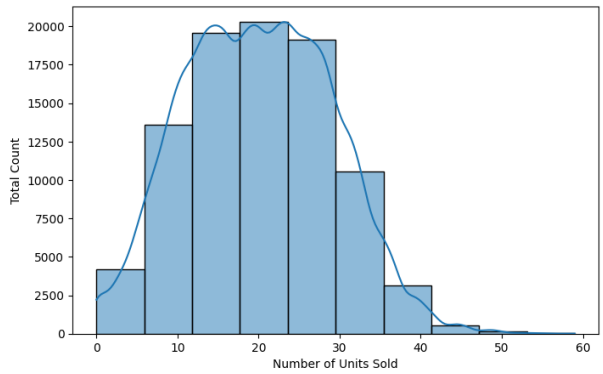
Entities:

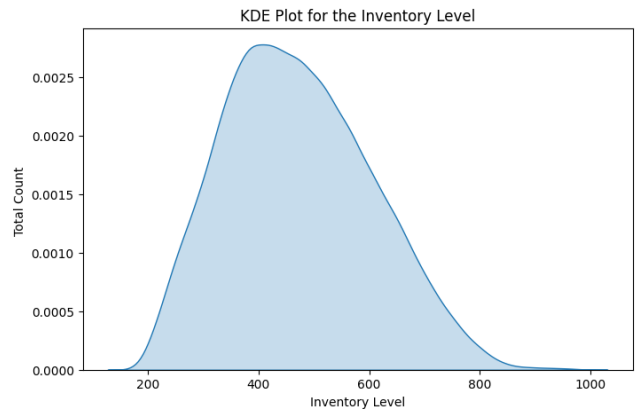
- 50 Products (SKU)
- 5 Warehouses
- 10 Suppliers
- 4 Regions

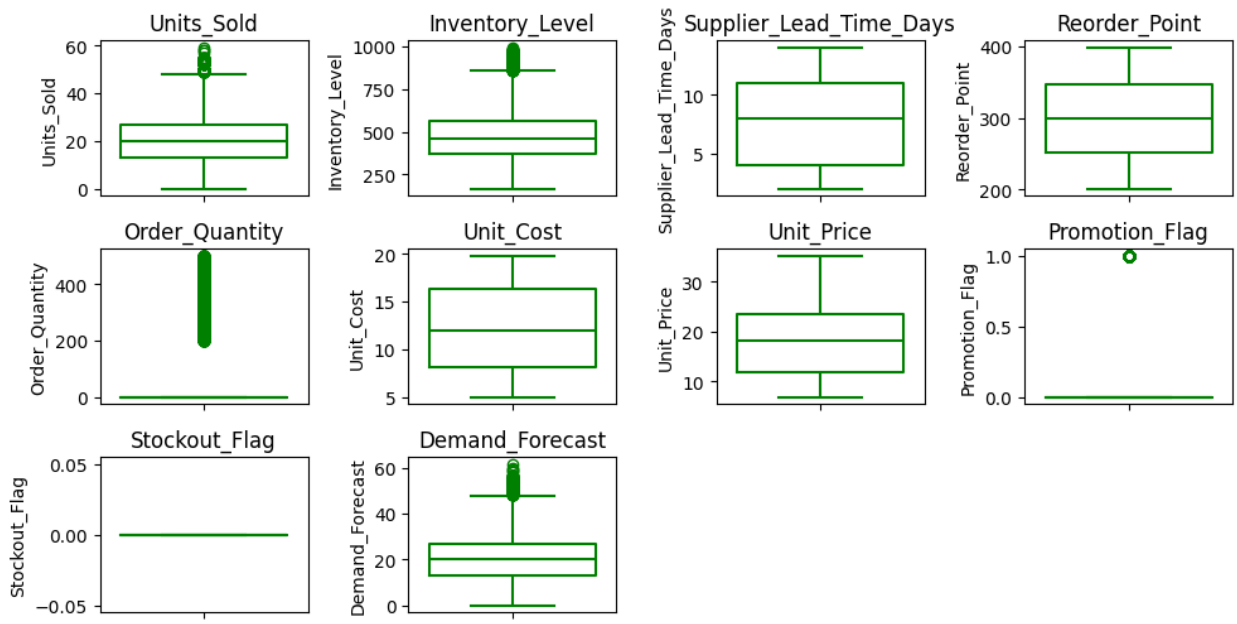
Data Quality Assessment

- No missing values
- No duplicate records

Univariate Analysis:

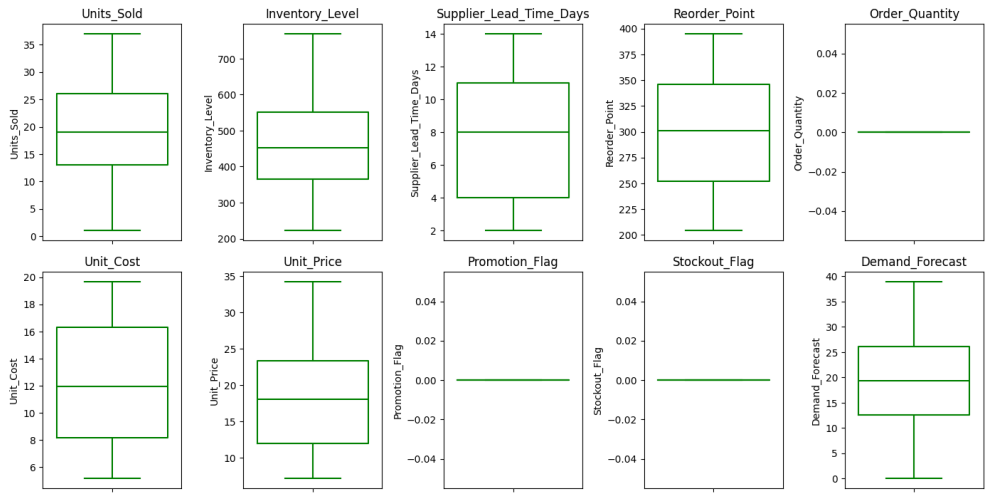






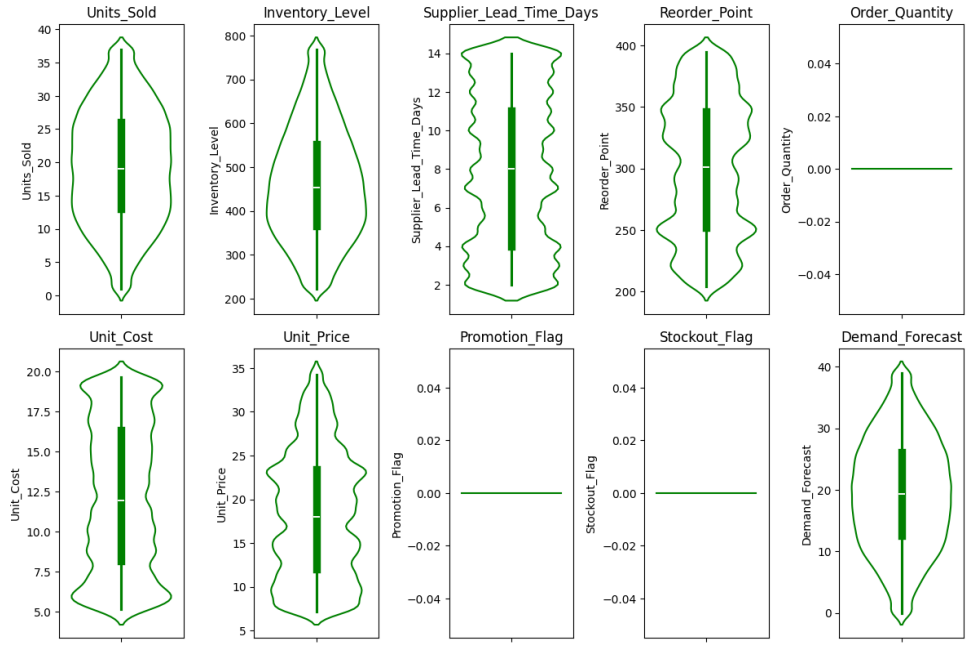
The univariate analysis was conducted to understand the distribution and characteristics of individual variables in the dataset. Most numerical variables, including Units Sold, Inventory Level, and Demand Forecast, exhibit approximately normal distributions with slight skewness in some cases. Inventory Level and Unit Price show mild right skewness, indicating a higher concentration of lower values with a few larger observations. Reorder Point is slightly left-skewed, suggesting consistency in replenishment strategy. Supplier Lead Time ranges between 2 to 14 days with moderate variability across suppliers. Overall, the dataset demonstrates stable distributions with no extreme anomalies after outlier treatment, making it suitable for further analysis.

EDA and Box Plot after IQR, and Capping:



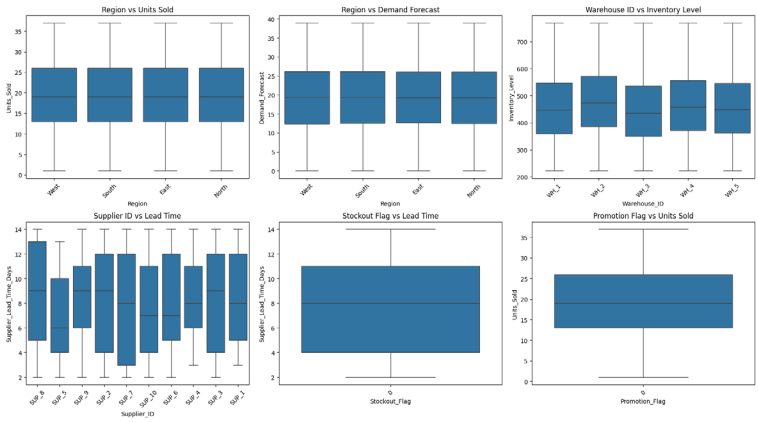
After applying capping, the boxplots indicate that extreme outliers have been effectively removed, and all variables now fall within a controlled range. The median values are well-centered within the interquartile range for most features, suggesting stable distributions. Variables such as Inventory Level and Demand Forecast show a relatively balanced spread, while Unit Cost and Unit Price exhibit moderate variability.

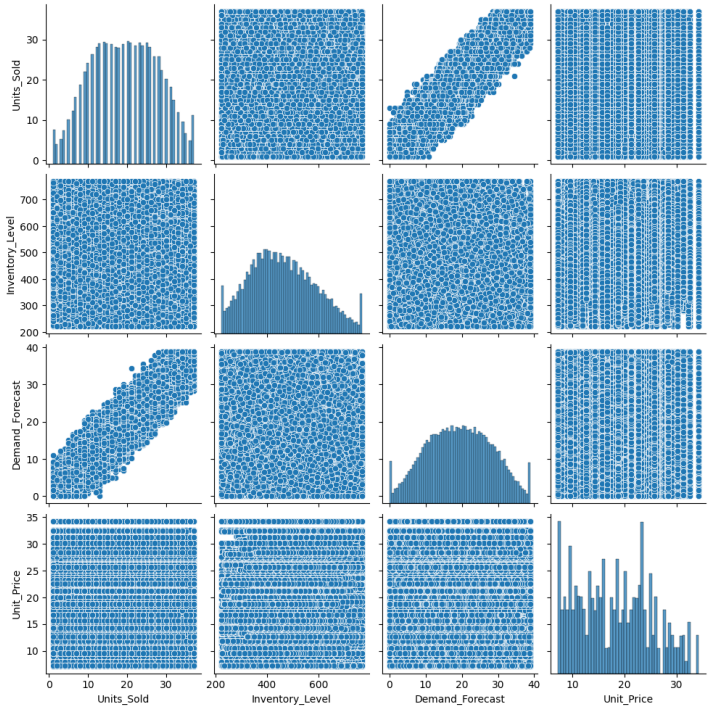
To check the spread we can see the violin plot to make sure the spread is okay.



The violin plots provide a detailed view of the distribution and density of numerical variables. Most features exhibit relatively symmetric distributions with moderate spread, while variables such as Inventory Level and Demand Forecast show consistent variability. Certain variables like Order Quantity, Promotion Flag, and Stockout Flag display no variation, indicating potential data limitations rather than meaningful business patterns.

Bivariate Analysis:

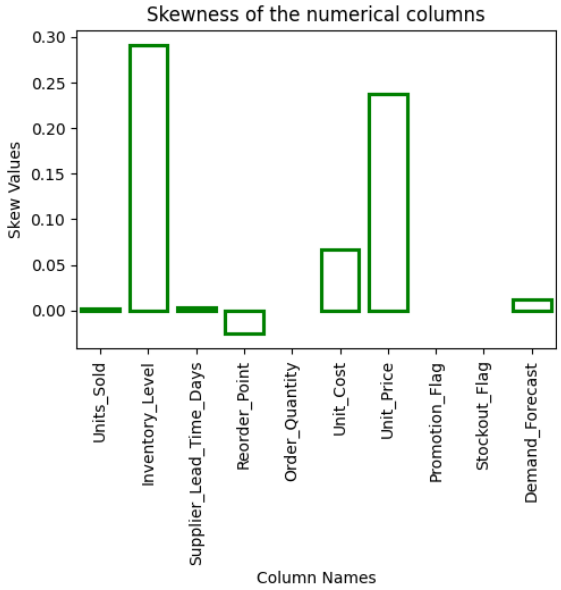


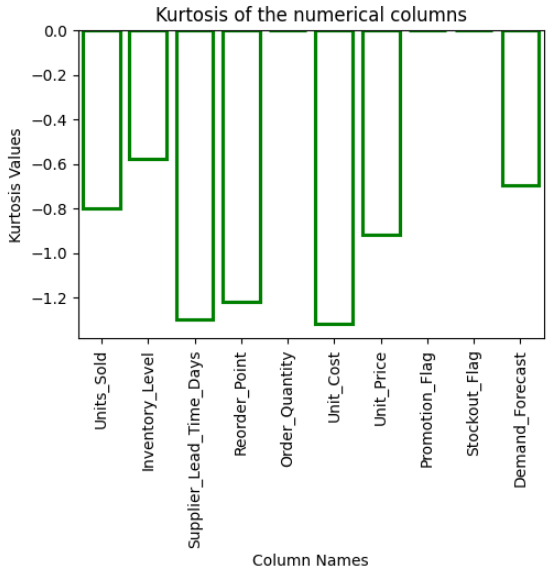


The bivariate analysis was conducted to examine the relationships between key supply chain variables using both scatter plots and categorical comparisons. A strong positive relationship is observed between Units Sold and Demand Forecast, indicating that the forecasting model closely aligns with actual sales and can be considered reliable for decision-making. In contrast, no clear relationship is found between Units Sold and Inventory Level or Unit Price, suggesting that inventory planning and pricing strategies are not directly influencing sales performance. A moderate positive relationship is identified between Inventory Level and Reorder Point, indicating that replenishment strategies are partially aligned with inventory levels.

From the categorical analysis, sales and demand distributions remain consistent across regions, showing no significant regional impact. However, inventory levels vary across warehouses, highlighting potential inefficiencies in stock distribution. Supplier Lead Time shows noticeable variation across suppliers, indicating inconsistent supplier performance, which may pose operational risks. Additionally, variables such as Stockout Flag, Promotion Flag, and Order Quantity show no meaningful variation, indicating data quality issues rather than valid business insights.

Skewness and Kurtosis Analysis:



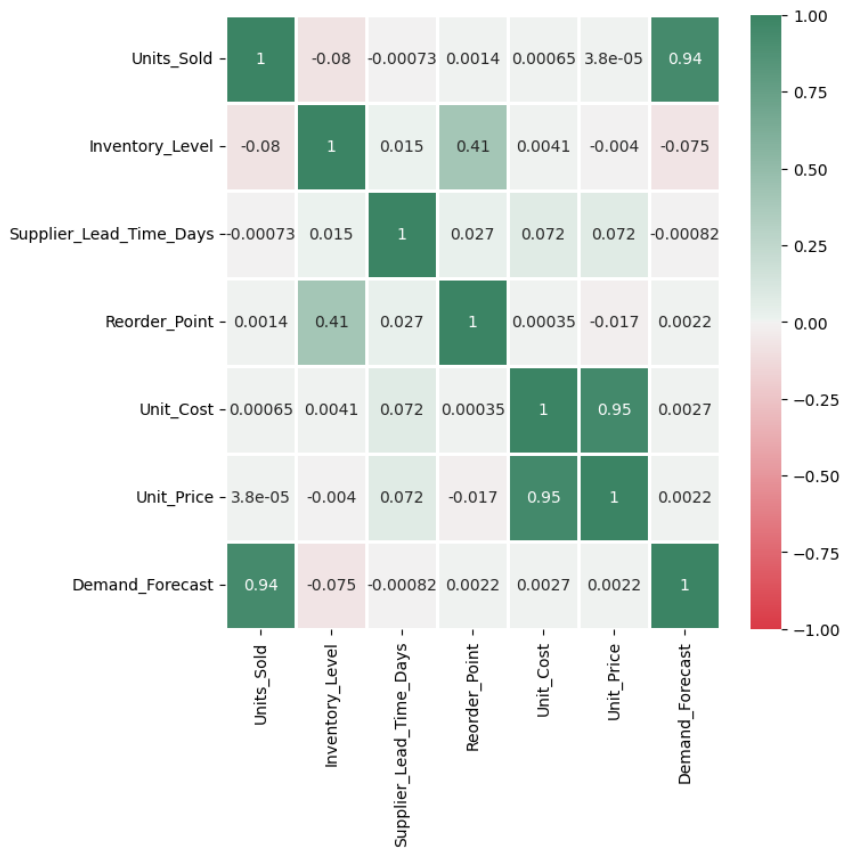


The skewness and kurtosis analysis was conducted to understand the distribution shape and tail behavior of the numerical variables. The skewness plot indicates that most variables are approximately symmetric, with values close to zero. However, slight positive skewness is observed in variables such as Inventory Level and Unit Price, suggesting a longer right tail where a few higher values exist. Reorder Point shows slight negative skewness, indicating a marginal concentration of higher values with a longer

tail on the lower side. Other variables, including Units Sold and Demand Forecast, exhibit near-zero skewness, implying balanced distributions.

The kurtosis plot reveals that most variables have negative kurtosis (platykurtic distribution), indicating that the data is flatter than a normal distribution with lighter tails and fewer extreme outliers. Variables such as Supplier Lead Time, Unit Cost, and Reorder Point show relatively lower kurtosis values, confirming the absence of heavy tails. Additionally, variables like Promotion Flag and Stockout Flag show zero or near-zero kurtosis due to lack of variability, reflecting data limitations rather than meaningful statistical behavior.

Correlation Analysis



The correlation analysis was performed to examine the linear relationships between key numerical variables in the dataset. The heatmap reveals a strong positive correlation between Units Sold and Demand Forecast (0.94), indicating that the demand forecasting model is highly aligned with actual sales and serves as a reliable predictor. Similarly, Unit Cost and Unit Price exhibit a very strong positive correlation (0.95), suggesting a consistent pricing strategy where prices are closely derived from costs.

A moderate positive relationship is observed between Inventory Level and Reorder Point (0.41), indicating that inventory replenishment decisions are partially influenced by stock levels. However, most other variables, including Units Sold with Inventory Level, Unit Price, and Supplier Lead Time, show weak or negligible correlations, suggesting limited direct impact on sales performance.

Additionally, Supplier Lead Time shows minimal correlation with other variables, indicating that supplier delays are not directly captured in sales or inventory behavior within this dataset.